A380X combined ELEC busbar and FUEL wing alerts

The A380X ECAM raises the FCOM combined busbar alerts, ELEC STATIC INV FAULT and the FUEL wing feed pump, wings balance and inner tank alerts, on top of the ELEC alerts and the A380X failures merged together

Aircraft	A380X
Where to find it	flyPad > Failures > ATA 24 and ATA 28; overhead ELEC and FUEL panels (CROSSFEED, feed pump pbs); E/WD, STATUS, SD ELEC and FUEL pages; T.O CONFIG key
Branch	`core/a380x-elec-fuel-alerts`, based on `core/a380x-elec` with `core/failures-a380x` merged in (stacked: needs both merged first; core/a380x-elec itself needs core/surveillance, core/a380x-mfd-fms and core/gps). Why a separate branch: the combined busbar and wing feed alerts need the ELEC alerts of core/a380x-elec and the FWS of core/failures-a380x, and merging either of those two into the other would put each scope into the other. The merge keeps both failure lists, both flyPad realism settings and the ECAM texts of both once, and applies the merge notes of core/failures-a380x: T.O CONFIG NORMAL reads the ELEC GEN / DRIVE alerts, ELEC EMER CONFIG reads the FWS signal inhibited by ALL ENG FLAME OUT; the FUEL JETTISON pbs are still read directly (no core/fuel here). Merge core/a380x-elec and core/failures-a380x into it again when they change. The features of the two parents are described in their own manuals.
Status	Tested in the sim by the user on the integration build: ELEC AC BUS 2+4 FAULT OK (2026-10-09/10 night flight, batch 3 test C1) and FUEL L WING FEED PMPs FAULT OK (2026-10-10 morning, batch 3 test W1). The other combined busbar alerts, STATIC INV FAULT, WINGS NOT BALANCED / BALANCED and INR TKs QTY LO: not sim-tested (unit tests FwsElecAlertsRound8.spec.ts, FwsFuelAlertsRound8.spec.ts). Integration: T.O CONFIG NORMAL with GEN 3 OFF recorded as passed (batch 2 test N3); no ELEC EMER CONFIG during ALL ENG FLAME OUT tested on develop 2026-10-06. The core branch itself was not sim-tested. A full MSFS restart is needed after installing.
Pictures	No pictures: the alerts use the stock E/WD, STATUS and SD pages.

1. Combined ELEC busbar alerts and ELEC STATIC INV FAULT

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 24; E/WD, STATUS, SD ELEC AC / DC | **Status:** ELEC AC BUS 2+4 FAULT tested in the sim by the user on the integration build: OK (2026-10-09/10 night flight, batch 3 test C1). The other combined alerts and STATIC INV FAULT: not sim-tested (unit tests FwsElecAlertsRound8.spec.ts).

What it is for

A380 FCOM PRO-ABN-ECAM-10-24 has combined alerts for the loss of several busbars together. The A380X showed the single busbar alerts side by side instead. Six combined alerts now trigger when all their busbars are lost and replace the single alerts they cover, with their procedure lines, flight phase inhibitions (phases 4, 5 and 10) and STATUS:

Alert	Busbars lost
ELEC AC BUS 1+2 & DC BUS 1 FAULT	AC 1, AC 2, DC 1
ELEC AC BUS 2+3 & DC BUS 1+2 FAULT	AC 2, AC 3, DC 1, DC 2
ELEC AC BUS 2+4 FAULT	AC 2, AC 4
ELEC AC BUS 3+4 & DC BUS 2 FAULT	AC 3, AC 4, DC 2
ELEC DC BUS 1+2 FAULT	DC 1, DC 2
ELEC DC BUS 1+ESS FAULT	DC 1, DC ESS

- A combined alert whose busbars are all part of another active one is replaced by it (DC BUS 1+2 FAULT inside AC BUS 2+3 & DC BUS 1+2 FAULT); combined alerts that only overlap are both shown.
- The fuel lines of these procedures depend on the flight time above FL 300 since departure (more or less than 30 min, counted from lift-off) and lead to gravity feeding (descent, CROSSFEED ... OFF, EXPECT NORM COLLECTOR CELL DEPLETION, AVOID NEG G LOAD), shown in flight only.
- The STATUS lists the FCOM lines that have an ECAM text, for example AC BUS 2+4 FAULT: MAX SPEED 310 KT, FUEL PARTLY UNUSABLE; INOP SYS F/CTL PROT, AP 1+2, A/THR, FD 1+2, ADR 2+3, TCAS 1+2, SATCOM, JETTISON, FEED TK 1 and 4 PMPs; INFO ALTN LAW : PROT LOST, FEED TK 1 (4) BY GRVTY ONLY. The JETTISON INOP SYS text (also in AC BUS 2+4, DC BUS 1+2 and DC BUS 1+ESS FAULT) is added here, identical to the one of `core/fuel`.
- **ELEC STATIC INV FAULT** (crew awareness, phases 3 to 10 inhibited): the static inverter runs only on batteries (no AC ESS power, both battery contactors closed); it counts as failed when it runs but gives no normal output.

Limitations and why

Limitation	Why	What to do
Some FCOM STATUS lines are missing (for example ATC COM VOICE ONLY, VHF 3, MOST FUEL XFR PMPs).	They have no ECAM text in FBW yet.	-
REDUND LOSS lines are not shown for these alerts.	They never reach the STATUS page in the FBW FWS.	-
The combined alerts do not trigger in ELEC EMER CONFIG.	Design choice: the bus fault alerts need one AC busbar powered; ELEC EMER CONFIG covers that case.	-
VENT AVNCS EXTRACT OVRD and OUTR TK XFR MAN lines are never ticked.	No avionics ventilation pb-sw; the manual outer tank transfer pb-sw is inoperative in the sim.	-

How to use

1. When a combined ELEC alert shows, follow its E/WD procedure (the sensed lines tick as you act).
2. Read the STATUS page; in flight follow the gravity feeding lines if they show.

How to test

From batch 3 test C1 (`reports/test-card-batch3-2026-10-10.md`). Setup: cruise above FL300, AP ON.

#	Do this	Expected result
1	flyPad > Failures > ATA 24 Electrical Power > **AC 2**, then **AC 4**.	Master caution and single chime; E/WD ELEC AC BUS 2+4 FAULT, not the single AC BUS 2 / AC BUS 4 alerts; SD ELEC AC page.
2	ECAM control panel: STS.	STATUS: ALTN LAW : PROT LOST, MAX SPEED 310 KT, INOP SYS among others JETTISON.
3	Deactivate both. Fail **DC 1**, then **DC 2**.	ELEC DC BUS 1+2 FAULT instead of DC BUS 1 / DC BUS 2 FAULT. Deactivate both.
4	On the ground, aircraft on batteries only (BAT pbs AUTO, no EXT PWR, no APU). Fail **Static Inverter**.	ELEC STATIC INV FAULT. Deactivate it.

2. FUEL wing feed pumps, wings balance and inner tank alerts

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28; overhead FUEL panel; E/WD, SD FUEL | **Status:** FUEL L WING FEED PMPs FAULT tested in the sim by the user on the integration build: OK (2026-10-10 morning, batch 3 test W1). WINGS NOT BALANCED / BALANCED and INR TKs QTY LO: not sim-tested (unit tests FwsFuelAlertsRound8.spec.ts).

What it is for

Four FUEL alerts of A380 FCOM PRO-ABN-ECAM-10-28 with their crossfeed procedure lines:

- **FUEL L (R) WING FEED PMPs FAULT**: all pumps of feed tanks 1 and 2 (left) or 3 and 4 (right) at low pressure, abnormally not running or switched off. It replaces the two FEED TK n MAIN + STBY PMPs FAULT alerts of its tanks. Inhibited in phases 4 to 7, 9 and 10. Pumps switched off on request of FUEL WINGS NOT BALANCED do not count.
- **FUEL WINGS NOT BALANCED**: on the ground, an imbalance above 6 600 lb between the symmetric feed tanks or the inner tanks, 5 500 lb between the mid tanks or 2 600 lb between the outer tanks (not while refuelling); in flight, more than 6 600 lb between feed tanks 1 and 4 or 2 and 3. In flight it stays until that pair is balanced (the lighter tank is no longer lighter), then **FUEL WINGS BALANCED** shows until its pumps are back ON and its crossfeeds OFF. Inhibited in phases 3 to 7, 9 and 10.
- **FUEL INR TKs QTY LO**: fuel on board above 110 200 lb with an inner tank below 11 200 lb (ZFW and ZFCG entered, both outer tanks full). Inhibited in phases 3 to 11.

Limitations and why

Limitation	Why	What to do
The "balanced" point of WINGS NOT BALANCED is a design choice.	The FCOM gives no balanced threshold.	-
Landing ends the in-flight balance procedures.	Design choice.	-

How to use

1. When a FUEL alert shows, follow its E/WD lines with the CROSSFEED and feed pump pbs of the overhead FUEL panel.
2. Watch the feed tank quantities on the SD FUEL page until the alert clears.

How to test

From batch 3 test W1 (`reports/test-card-batch3-2026-10-10.md`). Setup: cruise.

#	Do this	Expected result
1	flyPad > Failures > ATA 28 Fuel > **Feed tank 1 main pump**, **Feed tank 1 standby pump**, **Feed tank 2 main pump** and **Feed tank 2 standby pump**.	FUEL L WING FEED PMPs FAULT replaces the two FEED TK MAIN + STBY PMPs alerts.
2	Follow the E/WD lines (CROSSFEED pbs on the overhead FUEL panel).	The crossfeed lines tick.
3	Deactivate the four failures; CROSSFEED pbs back to normal.	The alert goes.
4	On the ground, engines running, set the left outer tank about 1.5 t (3 300 lb) above the right one (fuel set per tank, for example with the MSFS fuel menu).	FUEL WINGS NOT BALANCED. Set the tanks equal: it goes.

3. ELEC alerts with the A380X failures: T.O CONFIG and ELEC EMER CONFIG

> **Aircraft:** A380X | **Where:** T.O CONFIG key; E/WD | **Status:** On the integration build (develop): T.O CONFIG NORMAL blocked by GEN 3 OFF recorded as passed in the commit message (batch 2 test N3); no ELEC EMER CONFIG during ALL ENG FLAME OUT tested OK on develop 2026-10-06 (S6 T1). The core branch itself was not sim-tested.

What it is for

What this branch adds by having both parents together:

- **T.O CONFIG NORMAL** (the A380 rule of `core/failures-a380x`) also checks the ELEC GEN 1 (2)(3)(4) FAULT, GEN OFF and DRIVE DISCONNECTED alerts of `core/a380x-elec`, as in the FCOM list (DSC-31-40-10).
- **ELEC EMER CONFIG** (of `core/a380x-elec`) is not shown while ENG ALL ENG FLAME OUT is (FCOM: "This alert inhibits the ELEC EMER CONFIG alert").

How to use

1. Before take-off press T.O CONFIG with all four generators ON.
2. In an all engines flame out, follow ALL ENG FLAME OUT; ELEC EMER CONFIG does not come on top of it.

How to test

#	Do this	Expected result
1	On the ground, four engines running, ready for take-off. Overhead ELEC panel: GEN 3 pb OFF. ECAM control panel: T.O CONFIG key.	ELEC GEN 3 OFF; no T.O CONFIG NORMAL.
2	GEN 3 pb ON, wait 5 s, T.O CONFIG key again.	T.O CONFIG NORMAL.
3	In flight at FL200, 300 kt, APU OFF: activate the four "Engine n flameout (crew relight possible)" failures (flyPad > Failures > ATA 72). Wait 20 s.	Red ENG ALL ENG FLAME OUT; the RAT deploys; no ELEC EMER CONFIG. Relight as in the core/failures-a380x manual.